Lecture 13

Boundaries, defects and non-invertible symmetries

Objectives. *Lectures 11 and 12 built module categories, dual categories and the centre as mathematics. This lecture interprets them physically. The toric code has rough and smooth lattice boundaries, corresponding to the two module categories over* $\mathrm{Vec}_{\mathbb{Z}_2}$. *The anyons absorbed at each boundary form a* Lagrangian algebra. *In the critical Ising field theory, an interface between the theory and its gauged version becomes a topological line* N. *Its fusion rule* $N \otimes N = 1 \oplus \eta$ *makes this symmetry non-invertible. Gauging a group symmetry means summing over networks of its symmetry lines. Full gauging uses a module category with one simple object, which exists only when the symmetry has no anomaly.*

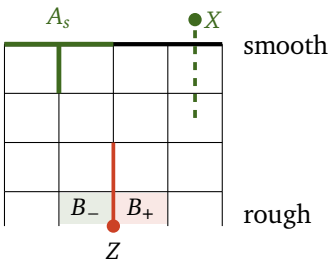
By the end of this lecture you will be able to:

- Find the anyons that condense on the rough and smooth boundaries of the toric code, from the lattice.
- Define commutative, connected, étale and Lagrangian algebras, and find the Lagrangian algebras of the toric code; state their correspondence with module categories.
- State the Kramers–Wannier line of the critical Ising model, its fusion rule and its category, and explain why the lattice operator is not a fusion category.
- State full gauging of Vec_G^{ω} symmetry, its anomaly and its dual symmetry $\mathrm{Rep}(G)$.

13.1 The toric-code boundaries

Recall the toric code of *Quantum Computing* (Lecture 14): qubits on the edges of a square lattice, the star term $A_s = \prod X$ on the four edges at a vertex and the plaquette term $B_p = \prod Z$ around a face. A string of Z's on a path violates the stars at its two ends, creating e's; a string of X's crossing a path on the dual lattice violates the plaquettes at its ends, creating m's. In Lecture 12 these are the simple objects $e = (0, \chi_-)$ and $m = (1, 1)$ of $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$. These pairs use additive group labels: 0 is the identity and 1 is the generator previously called s.

A Hamiltonian is *gapped* if the energy gap above its ground states is bounded below by a constant independent of the system size (Kitaev and Kong, fn. 1). A boundary ends the lattice along a line, and its terms are the truncated stars and plaquettes that remain; it is a *gapped boundary* if the Hamiltonian with it stays gapped. There are two natural ways to cut:



The thick green edges support one boundary star. The shaded three-edge plaquettes B_- and B_+ share the dangling edge carrying the Z-string endpoint. The dashed X-string exits through the smooth boundary. At the *smooth* boundary (top) the lattice ends along a row of edges, each in one plaquette, and the boundary stars have three edges. At the *rough* boundary (bottom) edges stick out, each meeting one vertex, and the boundary plaquettes have three edges (Kitaev–Kong §2; Bravyi and Kitaev call the rough one an x-boundary and the smooth one a z-boundary, and the names vary between sources).

Theorem 13.1 (The toric-code boundaries)**.** *At the smooth boundary m condenses: an X-string ending on the boundary leaves no excitation there, while e cannot be absorbed. At the rough boundary e condenses and m cannot be absorbed. The excitations that remain are $\{1, e\}$ on the smooth boundary and $\{1, m\}$ on the rough one.*

Proof. Absorption is local: consider operators supported near one long boundary, away from its ends and from the other boundary. An X on an edge flips the plaquettes containing it, a Z flips the stars at its ends, and every operator is a sum of products of these. A smooth-boundary edge lies in one plaquette, so an X-string on the dual lattice that ends by crossing it violates only the plaquette at its other end: m is absorbed. But every edge there still has two ends, each with a star, so every product of Paulis supported there flips an even number of stars: the number of e's keeps its parity, and a single e cannot disappear. At the rough boundary the outer end of a dangling edge has no star, so a Z-string ending there violates nothing at that end: e is absorbed. Every edge, the dangling ones included, lies in two plaquettes, so the same parity count keeps a single m. ∎

So the two boundaries are told apart by which anyon disappears into them: m on the smooth one, e on the rough one (Kong–Zhang Rem. 4.1.1 calls them the m- and e-condensed boundaries). Bringing any bulk anyon to the boundary gives a boundary excitation, and fusion decides which. At the smooth boundary m becomes 1, so $\varepsilon = e \otimes m$ becomes e: the map is $1, m \mapsto 1$ and $e, \varepsilon \mapsto e$. At the rough boundary it is $1, e \mapsto 1$ and $m, \varepsilon \mapsto m$ (Kitaev–Kong, maps (3)–(4)). The boundary excitations fuse like $\mathbb{Z}_2$, but they are not anyons: on a line they cannot braid.

13.2 Module categories and boundaries; Lagrangian algebras

The anyons that condense on a boundary form an algebra in the braided centre, and the algebra must be of a special kind.

Definition 13.2 (Commutative, connected, étale, Lagrangian)**.** An algebra (L, μ, ι) in a braided category is *commutative* if $\mu \circ c_{L,L} = \mu$, and *connected* if $\dim \mathrm{Hom}(1, L) = 1$; it is *étale* if it is commutative and separable (Lecture 11). In a non-degenerate braided fusion category $\mathcal{D}$, a connected étale algebra L is *Lagrangian* if $\mathrm{FPdim}(L)^2 = \mathrm{FPdim}(\mathcal{D})$ (DMNO Defs. 3.1 and 4.6). For unitary categories, FPdim = d, so this is $d(L)^2 = \dim \mathcal{D}$.

Theorem 13.3 (Lagrangian algebras of the toric code · *proved given coherence and EGNO Prop. 7.8.30*)**.** *In $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ the Lagrangian algebras are exactly $1 \oplus e$ and $1 \oplus m$, up to isomorphism. The square-zero algebras on $1 \oplus e$, $1 \oplus m$ and $1 \oplus \varepsilon$ are connected, commutative and two-dimensional, but not separable.*

Proof. $\dim \mathcal{Z} = 4$, so $d(L) = 2$, and connectedness leaves $L = 1 \oplus X$ with $X \neq 1$ one of e, m, ε, each with $X \otimes X = 1$. The unit is 1; the product on $X \otimes X \to 1$ is a scalar λ, and associativity holds for every λ. If $\lambda = 0$, the copy of X is a submodule of the right module L with no complement, since $1 \cdot X = X$: the modules are not semisimple, so L is not separable (Lecture 11, EGNO Prop. 7.8.30); commutativity holds, both sides being 0. If $\lambda \neq 0$, rescaling X makes $\lambda = 1$, a unique algebra. It is separable: the map $\Delta : L \to L \otimes L$ sending 1 to $\frac{1}{2}(1 \otimes 1 + X \otimes X)$ and X to $\frac{1}{2}(X \otimes 1 + 1 \otimes X)$ is a bimodule section of μ, as for $\mathbb{C}[\mathbb{Z}_2]$, and a morphism of $\mathcal{Z}$, since each component maps between copies of one simple object. On $X \otimes X \cong 1$ the braiding $c_{X,X}$ acts by θ_X (Theorem 12.8), so $\mu \circ c = \mu$ forces $\theta_X = 1$: $X = e$ or m, not ε, whose $\theta = -1$. ∎

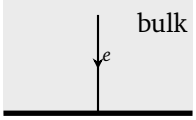
Two Lagrangian algebras and, in Lecture 11, two indecomposable module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$ are related by the following correspondence.

Theorem 13.4 (Davydov, Müger, Nikshych, Ostrik · *cited*)**.** *For a fusion category C, Lagrangian algebras in $\mathcal{Z}(C)$ correspond bijectively to indecomposable C-module categories $\mathcal{M}$; the category of L-modules in $\mathcal{Z}(C)$ is then $(C^*_{\mathcal{M}})^{\otimes \mathrm{op}}$, the dual with its tensor product reversed (DMNO Prop. 4.8 and Rem. 4.9(i), for right module categories; Kong 2014, (4.3)).*

For $\mathrm{Vec}_{\mathbb{Z}_2}$, the regular module ($H = 1$) has dual $\mathrm{Vec}_{\mathbb{Z}_2}$, and the module $\mathrm{Vect}_{\mathrm{fd}}$ ($H = \mathbb{Z}_2$) has dual $\mathrm{Rep}(\mathbb{Z}_2)$ (Theorem 12.2). The algebra $1 \oplus e$ condenses pure charges, the $H = 1$ boundary, with excitations $\{1, m\} \simeq \mathrm{Vec}_{\mathbb{Z}_2}$: the rough boundary. The algebra $1 \oplus m$ is the $H = \mathbb{Z}_2$ boundary, with $\{1, e\} \simeq \mathrm{Rep}(\mathbb{Z}_2)$: the smooth one (Beigi, Shor and Whalen, p. 13; Kong–Zhang Ex. 4.2.17). So the Morita equivalent symmetries $\mathrm{Vec}_{\mathbb{Z}_2}$ and $\mathrm{Rep}(\mathbb{Z}_2)$ of Lecture 12 live on two boundaries of one bulk. The toric code is the simplest *Levin–Wen* (string-net) model, the one built from $\mathrm{Vec}_{\mathbb{Z}_2}$ (Kitaev and Kong, §3). Its edges carry simple-object labels, and its local terms enforce fusion rules and F-moves. § 14.4 gives the construction.

Theorem 13.5 (Kitaev and Kong · *cited*)**.** *For a unitary fusion category C and an indecomposable C-module category $\mathcal{M}$, the Levin–Wen model built from C has a gapped boundary determined by $\mathcal{M}$, whose excitations form $\mathrm{Fun}_C(\mathcal{M}, \mathcal{M})$ (Kitaev and Kong, §4, with the tensor product in the reversed order; Kong 2014, §4.2).*

That every gapped boundary arises this way is a physical principle · *physical principle*: Kitaev and Kong "cannot readily defend the word 'all'" (p. 2), and a gapped edge needs a condensable set of bosons (Simon §25.8, pp. 359–360). With it, the gapped boundaries of the toric code are exactly the rough and the smooth one. The same count applies to any centre: $\mathcal{Z}(\mathrm{Fib})$ has exactly one gapped boundary, and $\mathcal{Z}(\mathrm{Vec}_G)$ for $G = S_3$ has four (Sheet 13). A boundary is drawn as a line ending the bulk, and an anyon in L ends on it:



rough boundary, $L = 1 \oplus e$

13.3 TFIM recap; KW at criticality; the continuum line and the lattice caveat

The transverse-field Ising chain on n sites, with periodic boundary conditions, is

$$H_{\text{chain}} = -\sum_j \left(Z_j Z_{j+1} + g\, X_j\right)$$

(Shao (3.41)). Its $\mathbb{Z}_2$ symmetry is $\eta = \prod_j X_j$. In *Integrability* (Lecture 18, eq. (18.1)) the same chain is written with $\sigma^x\sigma^x$ and σ^z, so $X \leftrightarrow \sigma^z$, $Z \leftrightarrow \sigma^x$, and η is its parity $\Pi = \prod \sigma^z$. For $g > 0$, in the thermodynamic limit, the ground state is unique and symmetric for $g > 1$, the symmetry is broken for $0 < g < 1$, and $g = 1$ is the critical point.

The *Kramers–Wannier* map sends $X_j \mapsto Z_{j-1}Z_j$ and $Z_{j-1}Z_j \mapsto X_{j-1}$: a site term becomes a bond term and a bond term a site term. Applied to H_{chain} it gives

$$-\sum_j \left(X_j + g\, Z_{j-1}Z_j\right) = g\, H_{\text{chain}}(1/g),$$

so $g \leftrightarrow 1/g$: the spectra at g and $1/g$ agree up to the scale g in the η-even sector of the periodic chain (*Integrability* Prop. 18.9). Away from $g = 1$ it maps the disordered phase to the ordered one: it is a duality between two Hamiltonians, not a symmetry of one (Shao, fn. 33). At $g = 1$ it commutes with H_{chain}, but it is not invertible. The product of all X_j is η, while the product of all $Z_{j-1}Z_j$ around the ring is 1; so the operator D realising the map has $D\eta = D$, and it kills the η-odd states.

A *topological line* in a field theory is an operator supported on a curve in spacetime whose small deformations change no correlation function away from the other insertions; along a spatial circle it commutes with the Hamiltonian (Shao §2.2).

To *gauge* a finite symmetry, we sum over insertions of networks of its lines, joined by the multiplication of an algebra in the symmetry category. The next section states when this construction is possible for the whole group.

Definition 13.6 (Tambara–Yamagami categories)**.** For a finite abelian group A, the fusion category $\mathrm{TY}(A, \chi, \pm 1/\sqrt{|A|})$ has simple objects $a \in A$ and one more, N, with $a \otimes b = ab$, $a \otimes N = N \otimes a = N$ and $N \otimes N = \bigoplus_{a\in A} a$. Its associators are fixed by a symmetric non-degenerate bicharacter $\chi : A \times A \to \mathrm{U}(1)$ and the sign of $\pm 1/\sqrt{|A|}$ (Bhardwaj and Tachikawa, (5.21) and after). A *bicharacter* is multiplicative in each variable:

$$\begin{gathered}\chi(ab, c) = \chi(a, c)\chi(b, c),\\ \chi(a, bc) = \chi(a, b)\chi(a, c)\\ (a, b, c \in A).\end{gathered}$$

It is symmetric if $\chi(a, b) = \chi(b, a)$, and non-degenerate if $\chi(a, b) = 1$ for every $b \in A$ forces $a = 1$. For $A = \mathbb{Z}_2 = \{1, \eta\}$, $\chi(\eta, \eta) = -1$.

Theorem 13.7 (The Kramers–Wannier line · *cited*)**.** *In the Ising conformal field theory, the continuum limit at $g = 1$, there is a topological line N with*

$$\begin{aligned}N \otimes N &= 1 \oplus \eta,\\ N \otimes \eta &= \eta \otimes N = N.\end{aligned}$$

The lines $1, \eta, N$ form the Tambara–Yamagami category $\mathrm{TY}(\mathbb{Z}_2, \chi, +1/\sqrt{2})$, the Ising fusion category with $\kappa = +1$ (Fröhlich, Fuchs, Runkel and Schweigert; Shao §3.1, (3.13); Bhardwaj and Tachikawa §5.6).

The parameter $+1/\sqrt{2}$ is the vacuum entry of the F-matrix in Theorem 6.7. Multiplying this entry by $d_\sigma = \sqrt{2}$ gives the Frobenius–Schur indicator $\kappa_\sigma = +1$. The negative parameter gives $\mathrm{SU}(2)_2$'s fusion category (Lecture 9). The line N is *non-invertible*: no line N' has $N \otimes N' = 1$.

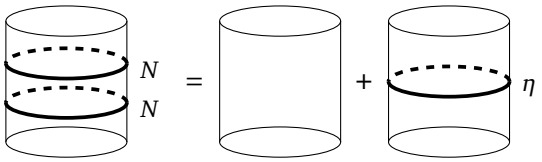
Gauging $\mathbb{Z}_2$ replaces its symmetry category by the dual $\mathrm{Rep}(\mathbb{Z}_2)$ of Example 12.3. At criticality the gauged theory is the same theory. Gauging η in half of space and not in the other half leaves an interface between the two, and at $g = 1$ the two halves are the same theory, so the interface is a topological line in one theory: this is N (Shao §6.3.1, (6.11)).

On the lattice the operator D realising the map does not obey this fusion rule. With the translation $\mathcal{T}$, $\mathcal{T} O_j \mathcal{T}^{-1} = O_{j+1}$,

$$D^2 = (1+\eta)\,\mathcal{T}^{-1},$$
$$\eta D = D\eta = D, \qquad DX_j = Z_{j-1}Z_j D$$

(Seiberg, Seifnashri and Shao, (1.4)–(1.5)). Together with $DZ_{j-1}Z_j = X_{j-1}D$, these relations fix D up to a scalar.[1]

The square of the lattice line contains a translation, which is not an internal symmetry: its defects add or remove sites, so the lattice symmetry mixes with space and is not a fusion category (SSS §1.2). Only the continuum limit, where $\mathcal{T} \to 1$ on the relevant states, gives $N \otimes N = 1 \oplus \eta$. Drawn on the cylinder of space and time, a line wraps the spatial circle, and in the continuum two parallel N lines are the sum of no line and an η line:



(13.1)

On the periodic chain the right side acquires $\mathcal{T}^{-1}$.

13.4 Full gauging, Rep(G), anomaly (statement)

Definition 13.8 (Full gauging, anomaly)**.** *Full gauging* uses the symmetry lines of the whole group in the network construction above. A fusion category is *anomalous* if it has no module category with one simple object (Thorngren and Wang, p. 13). By Theorem 11.6 with $H = G$, Vec_G^ω is anomalous exactly when $[\omega] \neq 0$.

Theorem 13.9 (Full gauging · *sketched*)**.** *Gauging all of a* Vec_G^ω *symmetry uses a rank-one module category, a* ψ *solving the module pentagon of Theorem 11.6 with* $H = G$*, which exists if and only if* $[\omega] = 0$*; the choices of* ψ *then differ by classes in* $H^2(G, \mathbb{C}^\times)$*. For* $\omega = \psi = 1$ *the gauged theory has the dual symmetry* $(\mathrm{Vec}_G)^*_{\mathrm{Vect_{fd}}} \simeq \mathrm{Rep}(G)$ *(Bhardwaj and Tachikawa §§4.2–4.6).*

[1] Comparing SSS (2.36) with Shao (3.46)–(3.48) gives

$$D_{\mathrm{Shao}} = D^\dagger/\sqrt{2}.$$

Thus Shao's (3.51) reads

$$D^2_{\mathrm{Shao}} = \frac{1}{2}(1+\eta)\mathcal{T}.$$

Sketch. Gauging inserts a fine network of symmetry lines, glued by an algebra in the symmetry category; for all of G this is $\bigoplus_g \delta_g$ with a product twisted by ψ, whose module category is the rank-one module (Lecture 11), and which is an algebra only when $[\omega] = 0$. The lines that remain topological in the gauged theory are the module endofunctors of that module category, its dual (Theorem 12.2). ∎

For $\mathbb{Z}_2$ this is Kramers–Wannier again: gauging η gives a theory with symmetry $\mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$. For a non-abelian group the dual is not a group at all: gauging S_3 gives the symmetry $\mathrm{Rep}(S_3)$, in which $\mathrm{std} \otimes \mathrm{std} \cong \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}$ (Lecture 3), so the line std has no inverse. Gauging an ordinary symmetry can produce a non-invertible one. The semion's $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ cannot be gauged: its only module category is the regular one. Gauging a subgroup uses the other module categories, beyond this course.

Summary.

- *Boundaries:* the toric code has the smooth (m condenses, excitations $\{1, e\}$) and the rough (e condenses, excitations $\{1, m\}$) lattice boundaries (Theorem 13.1); by the physical principle they are its only gapped ones.
- *Lagrangian algebras:* connected étale L with $d(L)^2 = \dim \mathcal{Z}$; in the toric code exactly $1 \oplus e$ and $1 \oplus m$ (Theorem 13.3); they match the module categories, and the excitations are the duals with the tensor product reversed (Theorems 13.4 and 13.5).
- *Non-invertible symmetry:* the critical Ising field theory has the line N with $N \otimes N = 1 \oplus \eta$, the category $\mathrm{TY}(\mathbb{Z}_2, \chi, +1/\sqrt{2})$ (Theorem 13.7); on the periodic chain $D^2 = (1 + \eta)\mathcal{T}^{-1}$ instead.
- *Gauging:* full gauging of $\mathrm{Vec}^{\omega}_{G}$ needs $[\omega] = 0$ and, for $\omega = \psi = 1$, yields $\mathrm{Rep}(G)$ (Theorem 13.9).
- *Next:* Lecture 14 assembles the course into topological quantum field theory.

References

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